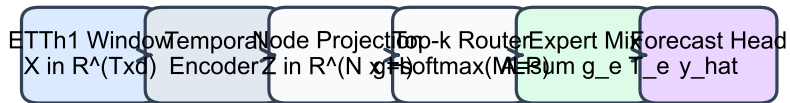


F2. ETTh1 Modeling Pipeline and Control Branches



$$\text{Eq(1) } p = \text{mean}_t(X_t) \quad \text{Eq(2) } z = \text{GRU}(p, N, H) \mid \text{Eq(3) } A = \text{sub_g_w_pred} * \text{pool}(\text{GraphOut})$$

Baseline branch (mean): MAE=2.234, latency=1.826 ms
 Control branch (GRU): MAE=1.438, latency=2.564 ms

Control axes: temporal encoder (mean vs GRU) and task regime (ETTh1-only vs dual-task), both validated with 10-seed runs.